



LESSON 6-3

Logarithms

Lesson Overview

Objective

Students will be able to:

- ✓ Understand the inverse relationship between exponents and logarithms.
- ✓ Use logarithms to solve exponential models.
- ✓ Evaluate logarithms using technology.

Essential Understanding

A logarithmic function is the inverse of an exponential function. Logarithms are found by determining the exponent that must be applied to a base to yield a given result.

Previously in this topic, students:

- Interpreted the key features of exponential functions.
- Used exponential functions to model relationships.

In this lesson, students:

- Learn about the relationship between exponents and logarithms and use this relationship to solve problems.
- Evaluate common and natural logarithms and solve equations involving logarithms.

Increasing Coherence

Examples 5 and 6 are not necessarily expected to be addressed on high stakes assessments.

Later in this topic students will:

- Evaluate logarithmic expressions using properties of logarithms.

This lesson emphasizes a blend of *conceptual understanding* and *procedural skill and fluency*.

- Students practice evaluating logarithms to understand the inverse relationship between exponents and logarithms.



Glossary is available at SavvasRealize.com.

Vocabulary Builder

REVIEW VOCABULARY

English | Spanish

- exponential function | *función exponencial*
- inverse function | *función inversa*

NEW VOCABULARY

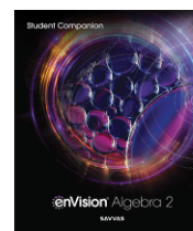
- common logarithm | *logaritmo común*
- logarithm | *logaritmo*
- logarithmic function | *función logarítmica*
- natural logarithmic function | *función logarítmica natural*

VOCABULARY ACTIVITY

Activate students' prior knowledge of *inverse functions* by having them find the inverses of two or three linear or quadratic functions. Then make the connection to the relationship between *exponential* and *logarithmic functions*.

Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



RIGOR



Mathematics Overview

Students understand the inverse relationship between exponents and logarithms and use what they know about this relationship to solve problems involving logarithms and exponents.

Students solve equations with logarithms and use logarithms to model and solve real-world problems. Students also use technology to evaluate common and natural logarithms.

Applying Math Practices

Reason Abstractly and Quantitatively

Students make sense of quantities and their relationships as they explore the inverse relationship between logarithms and exponents and as they evaluate common and natural logarithms.

Look For and Make Use of Structure

Students look for overall structure and patterns in exponents and logarithms, as well as apply general math rules to evaluate logarithms.

STEP 1 Explore



Activity

CRITIQUE & EXPLAIN

INSTRUCTIONAL FOCUS Students use their knowledge of powers of ten to determine a method for solving an exponential equation.

STUDENT COMPANION Students can complete the *Critique & Explain* activity in their *Student Companion*.

Before WHOLE CLASS

Implement Tasks That Promote Reasoning and Problem Solving **ETP**

- Q:** What do you notice about the relationship between the magnitude and the amplitude of an earthquake?
[As x increases by 1, y increases by a power of 10.]

During SMALL GROUP

Support Productive Struggle in Learning Mathematics **ETP**

- Q:** How are Taylor and Chen's work alike?
[Both students made estimates and both may have initially thought that x would be halfway between 3 and 4. However, Chen used the equation to check and refine his answer.]
- Q:** When would each student's method of solving the problem be more appropriate?
[Use Taylor's method when an estimate is acceptable. Use Chen's method when a more precise answer is needed.]

For Early Finishers

- Q:** Use the method you prefer to find the magnitude of an earthquake with amplitude 400.
[$10^{2.4} \approx 251$, $10^{2.5} \approx 316$, $10^{2.6} \approx 398$; The magnitude of an earthquake with amplitude 400 is about 2.6.]

After WHOLE CLASS

Facilitate Meaningful Mathematical Discourse **ETP**

Facilitate discussion by asking students to share ideas of how to find the exact value of the desired magnitude

- Q:** How would you solve the equation $y = 10^x$?
[Substitute 5,500 for y and solve for x .]
- Q:** Can you apply the same method to solve this equation? Explain.
[No; $y = 10^x$ cannot be solved by dividing both sides by 10 because x is an exponent, not a factor.]

Use with CRITIQUE & EXPLAIN

HABITS OF MIND

Reason Taylor reasoned that since 5,500 was halfway between 1,000 and 10,000 that the magnitude had to be halfway between 3 and 4. What is incorrect about Taylor's reasoning?

[Exponential functions increase exponentially, so they do not increase at a constant rate. There is more growth between $10^{3.5}$ and 10^4 than between 10^3 and $10^{3.5}$.]

Critique & Explain is available at SavvasRealize.com.

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CRITIQUE & EXPLAIN

Earthquakes make seismic waves through the ground. The equation $y = 10^x$ relates the height, or amplitude, in microns, of a seismic wave, y , and the power, or magnitude, x , of the ground-shaking it can cause.

Magnitude, x	Amplitude, y
2	100
3	1,000
?	5,500
4	10,000

Taylor and Chen used different methods to find the magnitude of the earthquake with amplitude 5,500.

Taylor

5,500 is halfway between 1,000 and 10,000.

3.5 is halfway between 3 and 4.

The magnitude is about 3.5.

Chen

$y = 10^x$

$10^3 = 1,000$

$10^4 = 10,000$

$10^{3.5} \approx 3,162$

$10^{3.7} \approx 5,012$

$10^{3.8} \approx 6,310$

$10^{3.9} \approx 7,943$

The magnitude is about 3.75.

A. What is the magnitude of an earthquake with amplitude 100,000? How do you know?

STUDENT EDITION

6-3 Logarithms

I CAN... evaluate and simplify logarithms.

VOCABULARY

- common logarithm
- logarithm
- logarithmic function
- natural logarithm

CRITIQUE & EXPLAIN

Earthquakes make seismic waves through the ground. The equation $y = 10^x$ relates the height, or amplitude, in microns on a seismograph, of a seismic wave, y , and the power, or magnitude, x , of the ground-shaking it can cause.

Magnitude, x	Amplitude, y
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The magnitude is about 3.75.

A. What is the magnitude of an earthquake with amplitude 100,000? How do you know?

B. **Construct Arguments** Critique Taylor's and Chen's work. Is each method valid? Could either method be improved?

C. Describe how to express the exact value of the desired magnitude.

What are logarithms and how are they used?

SAMPLE STUDENT WORK

- A. An earthquake with amplitude 100,000 has a magnitude of 5 because $10^5 = 100,000$.
- B. Taylor's method gives a rough estimate, but it is not valid if you need to find a more precise value. Chen's method gives a much closer approximation, although guess and check is not the most efficient way to solve problems.
- C. The exact value of the magnitude is the value of x when $10^x = 5500$.



INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning ETP

Remind students that they know how to find the inverses of some operations, but they have not yet learned how to find the inverse of exponentiation. They will study logarithms as the inverse of exponentiation and use this knowledge to interpret formulas involving logarithms.

EXAMPLE 1 Understand Logarithms

Establish Mathematics Goals to Focus Learning ETP

Q: How is solving $2^x = 8$ similar to solving $2x = 8$? different?

[Similar: both use inverse operations; different: inverse operation used to solve $2x = 8$ is division and inverse operation used to solve $2^x = 8$ is a logarithm.]

Q: You can calculate the value of x in $2^x = 8$ without rewriting the function as $\log_2 8 = x$. When might it be necessary to use this notation?

[It will be necessary to use the inverse form of an exponential function to solve more difficult problems.]

Try It! Answer

1. $x = \log_2 y$

Examples are available at SavvasRealize.com.

CONCEPTUAL UNDERSTANDING

USE STRUCTURE
Creating the notation $\log_2 x$ to represent the exponent to which you raise 2 to get x is similar to creating the radical notation $\sqrt[n]{x}$ to represent one number you can square to get x .

ESSENTIAL QUESTION What are logarithms and how are they evaluated?

EXAMPLE 1 Understand Logarithms

Solve $2^x = 8$.

The operation in $2^x = 8$ is exponentiation. To solve this equation, you need an inverse operation that answers the question, "To what exponent would you raise the base 2 to get 8?"

A logarithm is the inverse of exponentiation. To solve the equation $2^x = 8$, you can write $\log_2 8 = x$. Solving this gives $\log_2 8 = 3$ because $2^3 = 8$.

This is read "logarithm base 2 of 8" or "log base 2 of 8."

CONTINUED ON THE NEXT PAGE

302 TOPIC 6 Exponential and Logarithmic Functions

Additional Examples

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ADDITIONAL EXAMPLE 2A **ESSENTIAL QUESTION**

What is the logarithmic form of $\left(\frac{1}{3}\right)^{-3} = 27$?

The base is $\frac{1}{3}$.
The exponent is -3 .
The result is 27.

$\log_{\frac{1}{3}} 27 = -3$

SOLUTION

Example 2A In this additional example, students write the logarithmic form of an equation in which the base is a fraction.

Q: Does the fact that the base is a fraction change how you write the equation in logarithmic form?

[No; The base of the logarithmic form will be a fraction instead of a whole number, but the structure remains the same.]

Additional Examples are available at SavvasRealize.com.

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ADDITIONAL EXAMPLE 5 **ESSENTIAL QUESTION**

Solve $4^x = 18$. Round to the nearest thousandth.

$6x = \log_4 18$
 $6x \approx 2.085$
 $x \approx 0.347$

SOLUTION

Example 5 In this additional example, students solve an equation in which the exponent has a coefficient.

Q: How would the solution be different if the 3 were a coefficient of e rather than a coefficient of x ?

[The first step would be to divide both sides of the equation by 3 before converting to logarithmic form.]



STEP 2 Understand & Apply

EXAMPLE 2

Convert Between Exponential and Logarithmic Forms

Use and Connect Mathematical Representations **ETP**

Q: Why might you find it necessary to convert from one representation of the relationship in an exponential or logarithmic equation to the other?

[Both representations include the same information. One form may be preferred when a problem asks you to solve for a variable.]

Q: What are the exponential and logarithmic equations that show how 2, 5, and 25 are related?

[Use 5 as the base, 2 as the exponent and 25 as the result; $5^2 = 25$; $\log_5 25 = 2$]

Try It! Answers

2. a. $\log_7 343 = 3$
b. $4^2 = 16$

Common Error

Try It! 3 Some students confuse the placement of the various numbers when converting to another form of the equation.

Instead of writing $2^5 = 32$ as $\log_2 32 = 5$, students may write the logarithmic form as $\log_2 5 = 32$.

Show different parts of the expression using colors or a diagram.

Use with EXAMPLES 1 & 2

HABITS OF MIND

Communicate Precisely Write a sentence to describe what the equation $\log_a b = c$ means.

[b is equal to a raised to the c th power.]



Examples are available at SavvasRealize.com.

EXAMPLE 1 CONTINUED

The **logarithm** base b of x is defined as follows.

$$\log_b x = y \text{ if and only if } b^y = x, \text{ for } b > 0, b \neq 1, \text{ and } x > 0.$$

The **logarithmic function** $y = \log_b x$ is the inverse of the exponential function $y = b^x$.

Try It! 1. Write the logarithmic form of $y = 8^x$.

CONCEPT Exponential and Logarithmic Forms

Exponential form shows that a **base** raised to an **exponent** equals the **result**.

$$a^b = c$$

Logarithmic form shows that the log of the **result** with the given **base** equals the **exponent**.

$$\log_a c = b$$

When written in logarithmic form, the number that was the result of the exponential equation is often called the argument.

EXAMPLE 2 Convert Between Exponential and Logarithmic Forms

A. What is the logarithmic form of $3^4 = 81$?

The **base** is 3, the **exponent** is 4, and the **result** is 81.

So, in logarithmic form,

$$3^4 = 81 \rightarrow \log_3 81 = 4.$$

The logarithmic form of $3^4 = 81$ is $\log_3 81 = 4$.

B. What is the exponential form of $\log_{10} 1,000 = 3$?

The base is 10, the exponent is 3, and the result (or argument) is 1,000.

So, in exponential form,

$$\log_{10} 1,000 = 3 \rightarrow 10^3 = 1,000.$$

The exponential form of $\log_{10} 1,000 = 3$ is $10^3 = 1,000$.

Try It! 2. a. What is the logarithmic form of $7^3 = 343$?
b. What is the exponential form of $\log_4 16 = 2$?

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Extend Student Thinking

USE WITH EXAMPLE 2 Give advanced students an extra challenge by asking them to rewrite exponential and logarithmic equations that contain variables.

Q: Rewrite $y = x^5$ in logarithmic form.
[$\log_x y = 5$]

Q: Rewrite $\log_3 (n + 2) = n - 5$.
[$3^{n-5} = n + 2$]

Q: What value of n makes the equation $\log_3 (n + 2) = n - 5$ true?
[7]



EXAMPLE 3 Evaluate Logarithms

Support Productive Struggle in Learning Mathematics ETP

Q: Must all parts of a logarithmic expression be positive?

[No; The base must be positive by definition. The input value must be positive because any positive number raised to an exponent yields a positive number. But the properties of exponents state that $a^{-b} = \frac{1}{a^b}$, so the value of the expression as a whole can be negative.]

Try It! Answers

3. a. -4 b. undefined c. 9

Elicit and Use Evidence of Student Thinking ETP

Q: How is Part a in the Try It different from Part B of the example?

[In the Try It, the argument is a fraction, but in the example, the base is a fraction.]

EXAMPLE 4 Evaluate Common and Natural Logarithms

Pose Purposeful Questions ETP

Q: Why can the exponential form of the expression be used to check your answers?

[The exponential form of the expression is the inverse of the logarithmic form. The exponential form can be used to check logarithmic form in the same way that multiplication can be used to check division.]

Try It! Answers

4. a. 2.5065 b. 7.1025 c. -0.7696

HABITS OF MIND

Reason For $\log x$ or $\ln x$ to be defined, what must be true of x ?

[$x > 0$]

Use with EXAMPLES 3 & 4

Support Struggling Students

USE WITH EXAMPLE 3 Some students may have difficulty understanding how thinking of a logarithmic expression in terms of an exponential equation can help them evaluate logarithms.

Q: What does *evaluate* mean?

[find the value of]

Q: Must an expression include a variable in order to be evaluated?

[No, numerical expressions can be evaluated.]



Examples are available at SavvasRealize.com.

EXAMPLE 3 Evaluate Logarithms

What is the value of each logarithmic expression?

A. $\log_5 125$ THINK: $5^? = 125$ Since $5^3 = 125$, $\log_5 125 = 3$.	B. $\log_2 16$ THINK: $(\frac{1}{2})^? = 16$ Since $(\frac{1}{2})^{-2} = 16$, $\log_2 16 = -2$.
C. $\log_3 0$ THINK: $3^? = 0$ There is no such power, so $\log_3 0$ is undefined.	D. $\log_2 2^8$ THINK: $2^? = 2^8$ Since $2^8 = 2^8$, $\log_2 2^8 = 8$.

Try It! 3. What is the value of each logarithmic expression?

- a. $\log_3 (\frac{1}{3})$ b. $\log_7 (-7)$ c. $\log_5 5^9$

CONCEPT Common Logarithms and Natural Logarithms

The base 10 logarithm is called the **common logarithm** and is written as $\log x$ with the base of 10 implied.

The base e logarithm is called the **natural logarithm** and is written as $\ln x$.

EXAMPLE 4 Evaluate Common and Natural Logarithms

What is the value of each logarithmic expression to the nearest ten-thousandth?

A. $\log 900$ $\log 900 \approx 2.9542$	B. $\ln e$ $\ln e = 1$	C. $\ln(-1.87)$ $\ln(-1.87) \rightarrow \text{ERROR}$	Evaluate the exponential forms to reinforce these inverse operations. $10^{2.9542} \approx 900$ $e^1 = e$ $e^{-1.87} \approx -1.87$
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There is no exponent to which e can be raised in order to get a negative number, so $\ln(-1.87)$ is undefined.

Try It! 4. What is the value of each logarithmic expression to the nearest ten-thousandth?

- a. $\log 321$ b. $\ln 1,215$ c. $\log 0.17$

STEP 2 Understand & Apply

EXAMPLE 5 Solve Equations With Logarithms

Facilitate Meaningful Mathematical Discourse ETP

Q: How is solving exponential and logarithmic equations similar to solving linear equations? How is it different?
[Solving any type of equation requires the use of inverse operations. To solve linear equations, you use the inverse operations of addition/subtraction and multiplication/division. To solve exponential equations, you use logarithms, and vice versa.]

Try It! Answers

5. a. 34

b. 0.079

EXAMPLE 6 Use Logarithms to Solve Problems

Pose Purposeful Questions ETP

Q: How does converting the form of the equation help to find the magnitude of the earthquake?
[The equation can be converted from exponential form to a linear equation containing a logarithm. The logarithm can be evaluated and the linear equation can be solved in two more steps to reveal the magnitude.]

Try It! Answer

6. magnitude 7.5

Elicit and Use Evidence of Student Thinking ETP

Q: Do you think the magnitude of the earthquake in the Try It will be greater than or less than the magnitude of the earthquake in the example?
[It will be greater because $1.8 \times 10^{23} > 4.2 \times 10^{20}$.]

HABITS OF MIND

Use with EXAMPLES 5 & 6

Reason How do logarithms help you solve an equation in which the variable is an exponent?

[It allows you to rewrite the expression in terms of the variable.]

Examples are available at SavvasRealize.com.

COMMON ERROR
Remember that 10 is not a coefficient, but a base. You cannot divide both sides by 10 and then add 1 to solve for x .

EXAMPLE 5 Solve Equations With Logarithms

What is the solution to each equation? Round to the nearest thousandth.

A. $25 = 10^{x-1}$

$25 = 10^{x-1}$

$\log 25 = \log 10^{x-1}$

$1 + \log 25 = x$

$2.398 \approx x$

B. $\ln(2x + 3) = 4$

$\ln(2x + 3) = 4$

$2x + 3 = e^4$

$2x + 3 \approx 54.598$

$2x \approx 51.598$

$x \approx 25.799$

Convert to logarithmic form.

Convert to exponential form.

Try It! 5. Solve each equation. Round to the nearest thousandth.

a. $\log(3x - 2) = 2$

b. $e^{x+2} = 8$

APPLICATION

EXAMPLE 6 Use Logarithms to Solve Problems

The seismic energy, x , in joules can be estimated based on the magnitude, m , of an earthquake by the formula $x = 10^{1.5m+12}$. What is the magnitude of an earthquake with a seismic energy of 4.2×10^{20} joules?

Formulate Substitute 4.2×10^{20} for x in the formula.

$4.2 \times 10^{20} = 10^{1.5m+12}$

Compute Solve the equation for m .

$4.2 \times 10^{20} = 10^{1.5m+12}$

$\log(4.2 \times 10^{20}) = 1.5m + 12$

$20.6 \approx 1.5m + 12$

$5.75 \approx m$

Interpret The magnitude of the earthquake is about 5.75.

Verify the answer: $10^{1.5(5.75)+12} \approx 4.2 \times 10^{20}$

Try It! 6. What is the magnitude of an earthquake with a seismic energy of 1.8×10^{23} joules?



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ELL English Language Learners (Use with EXAMPLE 6)

READING BEGINNING Explain to students that the word *joules* is a homophone. Homophones are words that have the same pronunciation but different meanings and spellings. Write the words *joules* and *jewels*, along with their respective definitions, on the board. Provide sentences in which students fill in the blank either with joules or jewels.

Q: There are 20 _____ in the king's crown. [jewels]

Q: A heavy-duty surge protector will protect all of your electronic devices from surges up to 5,500 _____. [joules]

LISTENING DEVELOPING Place some small objects on a desk and use a magnifying glass to look at them.

Q: What is a magnifying glass used for? [to make small objects look larger]

Q: What do the words *magnify*, *magnificent*, and *magnitude* have in common? [the prefix magni-]

Q: What do you think the prefix *magni-* means, and how does it help you to understand the meaning of the word *magnitude*? [large; The magnitude of an earthquake is a measure of how large it is]

WRITING EXPANDING Ask students to write the definitions of *seismic energy*, *joules*, and *magnitude*. Then have them draw a diagram to show the connections between the words.

Q: Explain how *seismic energy* is related to *joules* and to *magnitude*.
[Seismic energy is measured in joules. There is a strong positive correlation between seismic energy and the magnitude of an earthquake.]

CONCEPT SUMMARY Logarithms

Q: Explain the relationship between logarithms and exponents. [Logarithms and exponents are inverse operations.]

Q: When is it useful to convert between exponential and logarithmic forms?

[It is useful to rewrite an exponential equation in logarithmic form when the exponent contains a variable. The logarithm can be evaluated and the resulting linear equation can easily be solved. Similarly, it is useful to rewrite a logarithmic equation in exponential form when the result (or argument) contains a variable. The exponent, with a base of 10 or e can be evaluated and the resulting linear equation can easily be solved.]

Do You UNDERSTAND? | Do You KNOW HOW?

Common Error

Exercise 11 Some students may think that e is a variable and not a constant. Explain that e is an irrational number called Euler's number. Compare e to π to help students understand that e represents a specific number that is approximately 2.718.

Answers

- Logarithms are exponents. Logarithms can be evaluated by rewriting them in exponential form.
- Amir possibly brought the negative sign outside the expression, which is not an equivalent step.
- Common logarithms use the base 10. Natural logarithms use the base e .
- You can convert the equation to logarithmic form, $t = \log 656$, which also solves the equation.
- $\log_2 \left(\frac{1}{64} \right) = -6$
- $\ln 54.6 \approx 4$
- $10^{2.301} \approx 200$
- $e^{3.22} \approx 25$
- 3
- 2
- 5
- $\ln \left(\frac{7}{4} \right)$, or 0.5596



Concept Summary, Do You Understand, and Do You Know How are available at SavvasRealize.com.

CONCEPT SUMMARY Logarithms

	Exponential Form		Logarithmic Form
ALGEBRA	$b^x = y$	$\longleftrightarrow$	$\log_b y = x$
WORDS	The base raised to the exponent is equal to a result.		The logarithm with a base b of the result (or argument) is equal to the exponent.
NUMBERS	$3^4 = 81$	$\longleftrightarrow$	$\log_3 81 = 4$

Do You UNDERSTAND?

- ESSENTIAL QUESTION** What are logarithms and how are they evaluated?
- Error Analysis** Amir said the expression $\log_5(-25)$ simplifies to -2. Explain Amir's possible error.
- Vocabulary** Explain the difference between the common logarithm and the natural logarithm.
- Make Sense and Persevere** How can logarithms help to solve an equation such as $10^t = 656$?

Do You KNOW HOW?

- Write each equation in logarithmic form.
- $2^{-6} = \frac{1}{64}$
 - $e^4 \approx 54.6$
- Write each equation in exponential form.
- $\log 200 \approx 2.301$
 - $\ln 25 \approx 3.22$
- Evaluate the expression.
- $\log_4 64$
 - $\log_3 \frac{1}{100}$
 - $\ln e^5$
 - Solve for x . $4e^x = 7$.

STEP 3 Practice & Problem Solving



PRACTICE & PROBLEM SOLVING



Practice & Problem Solving and video tutorials are available at SavvasRealize.com.

Scan the QR code to access a video tutorial.

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: Lesson Practice

Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
13–42, 47–57	13–20, 25–57

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply 2 points each

Practice 1 point each

Assessment Practice 1 point each

Performance Task 3 points

Item Analysis

Example	Items	DOK
1	19–22	1
2	23–30	1
3	31–38	1
	16, 55, 56	2
	17	3
4	39–44	1
	13, 15, 18	3
5	14, 45–50	2
6	51–54	2
	57	3

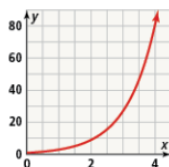
PRACTICE & PROBLEM SOLVING

UNDERSTAND

13. **Make Sense and Persevere** If the LN button on your calculator were broken, how could you still use your calculator to find the value of the expression $\ln 65$?
14. **Error Analysis** Describe and correct the error a student made in solving an exponential equation.

$$\begin{aligned} 16e^t &= 98 \\ e^t &= 6.125 \\ 6.125t &= \ln e \\ t &= \frac{\ln e}{6.125} \end{aligned}$$

15. **Higher Order Thinking** Use the graph of $y = 3^x$ to estimate the value of $\log_3 50$. Explain your reasoning.



16. **Generalize** For what values of x is the expression $\log_4 x < 0$ true?
17. **Use Structure** A student says that $\log_3\left(\frac{1}{27}\right)$ simplifies to -3 . Is the student correct? Explain.
18. **Use Structure** Explain why the expression $\ln 1,000$ is not equal to 3.
19. **Construct Arguments** Explain why $\log_b b^x = x$. Explain why $e^{\ln x} = x$.

PRACTICE

Write the logarithmic form of each exponential equation. SEE EXAMPLE 1

20. $y = 4^x$ 21. $y = 10^x$

22. $y = 7^x$ 23. $y = a^x$

Write each equation in logarithmic form. SEE EXAMPLE 2

24. $3^8 = 6,561$ 25. $e^{-3} \approx 0.0498$

26. $5^0 = 1$ 27. $7^3 = 343$

Write each equation in exponential form. SEE EXAMPLE 2

28. $\log_{100} \frac{1}{100} = -2$ 29. $\log_8 64 = 2$

30. $\ln 148.41 \approx 5$ 31. $\log_2 \frac{1}{32} = -5$

Evaluate each logarithmic expression. SEE EXAMPLE 3

32. $\log_5 \frac{1}{125} = -3$ 33. $\log_6(-216)$ undefined

34. $\log_3 3^4 = 4$ 35. $\log_2 32 = 5$

36. $\log_9 729 = 3$ 37. $\log_8 \frac{1}{64} = -2$

38. $\log_7 0$ undefined 39. $\log_7 7^a = a$

Use a calculator to evaluate each expression. Round to the nearest ten-thousandth. SEE EXAMPLE 4

40. $\log 78.5$ 41. $\log 0.24$

42. $\ln(-37)$ undefined 43. $\ln 41.5$

44. $\log 12$ 45. $\ln 3$

Solve each equation. Round answers to the nearest ten-thousandth. SEE EXAMPLES 5 AND 6

46. $\log(7x + 6) = 3$ 47. $2.75e^t = 38.6$

48. $\ln(3x - 1) = 2$ 49. $10^{t+1} = 50$

50. $1.5e^t = 27$ 51. $\log(x - 3) = -1$

52. How long does it take for \$250 to grow to \$600 at 4% annual percentage rate compounded continuously? Round to the nearest year.

LESSON 6-3 Logarithms 307

Answers

13. Graph $y = e^x$ and use the graph to find an approximate value of x for which y is equal to 65.
14. The student did not convert to logarithmic form correctly. The solution should be $t = \ln 6.125$.
15. 3.5 ; $\log_3 50$ is an expression equivalent to $3^x = 50$, so you need to find the value of x for which $y = 50$.
16. $\{x \mid 0 < x < 1\}$

17. yes; $3^{-3} = \frac{1}{27}$

18. Because the base is e and e^3 is not equal to 1,000.

19. $y = \log_b x$ and $y = b^x$ are inverses. Part of the definition of inverses is that $f(g(x)) = x = g(f(x))$. It follows that $\log_b b^x = x$. Since $y = e^x$ and $y = \ln x$ are inverses, then the same reasoning applies for $e^{\ln x} = x$.

20. $\log_4 y = x$

21. $\log y = x$

See answers for 22–31, 52 on the next page.



Answers

22. $\log_7 y = x$
23. $\log_a y = x$
24. $\log_3 6561 = 8$
25. $\ln 0.0498 \approx -3$
26. $\log_5 1 = 0$
27. $\log_7 343 = 3$
28. $10^{-2} = \frac{1}{100}$
29. $8^2 = 64$
30. $e^5 = 148.41$
31. $2^{-5} = \frac{1}{32}$
52. 22 years
53. Peter's account
54. a. magnitude 7.9
b. 31,009,857,353 kWh
c. 16,705 kWh
55. a. 183°F
b. 19 minutes
58. a. Account 1: $A = 400e^{0.08t}$; Account 2: $A = 500 + 75t$
b. yes; Account 2 is increasing at a constant rate while Account 1 is increasing at an increasing rate, so after a certain point the amount of money in Account 1 will exceed the amount of money in Account 2; in about 20 years.



Practice & Problem Solving is available at SavvasRealize.com.

PRACTICE & PROBLEM SOLVING

APPLY

53. **Model with Mathematics** Michael invests \$1,000 in an account that earns a 4.75% annual percentage rate compounded continuously. Peter invests \$1,200 in an account that earns a 4.25% annual percentage rate compounded continuously. Which person's account will grow to \$1,800 first?
54. **Reason** The Richter magnitude of an earthquake is $R = 0.67 \log(0.37E) + 1.46$, where E is the energy (in kilowatt-hours) released by the earthquake.
 - a. What is the magnitude of an earthquake that releases 11,800,000,000 kilowatt-hours of energy? Round to the nearest tenth.
 - b. How many kilowatt-hours of energy would an earthquake have to release in order to be an 8.2 on the Richter scale? Round to the nearest whole number.
 - c. What number of kilowatt-hours of energy would an earthquake have to release in order for walls to crack? Round to the nearest whole number.

At a Richter magnitude of 4 and above, the walls in your house may start to crack.



55. **Reason** The function $c(t) = 108e^{-0.08t} + 75$ calculates the temperature, in degrees Fahrenheit, of a cup of coffee that was handed out a drive-thru window t minutes ago.
 - a. What is the temperature of the coffee in the instant that it is handed out the window?
 - b. After how many minutes is the coffee in the cup 98 degrees Fahrenheit? Round to the nearest whole minute.

308 TOPIC 6 Exponential and Logarithmic Functions

ASSESSMENT PRACTICE

56. Sheba invests \$500 in an account that earns 2.5% annual percentage rate compounded continuously. How long will it take for her account to grow to \$700? **about $13\frac{1}{2}$ years**
57. **SAT/ACT** In the equation $\log_3 a = b$, if b is a whole number, which of the following CANNOT be a value for a ?
Ⓐ 1 Ⓑ 3 Ⓒ 6 Ⓓ 9 Ⓔ 81
58. **Performance Task** Money is deposited into two separate accounts. The money in one account is compounded continuously. The money in the other account is not compounded continuously. Neither account has any money withdrawn in the first 6 years.

Year	Account 1 Balance (\$)	Account 2 Balance (\$)
0	400	500
1	433.31	575
2	469.40	650
3	508.50	725
4	550.85	800
5	596.72	875

Part A Write a function to calculate the amount of money in each account given t , the number of years since the account was opened. Describe the growth in each account.

Part B Will the amount of money in Account 1 ever exceed the amount of money in Account 2? Explain. If so, when will that occur?



STEP 4 Assess & Differentiate

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.

Students can take the Lesson Quiz online or you can download a printable copy from [SavvasRealize.com](https://www.savvasrealize.com). The Lesson Quiz is also available in the *Assessment Resources* book.

Item Analysis

Item	DOK	Skills Review and Practice
1	2	F69
2	2	F73
3	2	F73
4	1	F69
5	2	F69



Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	- Reteach to Build Understanding - Mathematical Literacy and Vocabulary - Lesson Virtual Nerd videos
O On-Level	4 points	Enrichment
A Advanced	5 points	Enrichment



Lesson Quiz is available at [SavvasRealize.com](https://www.savvasrealize.com).

ASSESSMENT RESOURCES

Name _____

enVision Algebra 2

SavvasRealize.com

6-3 Lesson Quiz

Logarithms

- Which function is the inverse of the exponential function $y = 3^x$?
 - ☐ A $y = x^3$
 - ☐ B $y = 3^x$
 - ☒ C $y = \log_3 x$
 - ☐ D $y = \log_x 3$
- Which is the best estimate of the value of $\ln 16$?
 - ☐ A 1.20
 - ☒ B 2.77
 - ☐ C 3.23
 - ☐ D 4.00
- Which is the logarithmic equation $\log_8 64 = a$, written in exponential form?
 - ☐ A $64^a = 8$
 - ☐ B $a^8 = 64$
 - ☐ C $a^{64} = 8$
 - ☒ D $8^a = 64$
- Select all the logarithmic expressions that have been evaluated correctly.
 - ☒ A. $\ln 1 = 0$
 - ☐ B. $\log_2 9 = 3$
 - ☐ C. $\log_{100} \frac{1}{100} = \frac{1}{2}$
 - ☐ D. $\log_3 (-1) = \frac{1}{3}$
 - ☒ E. $\log_5 \frac{1}{125} = -3$
- What is the solution to the equation $\log(2x + 4) = 2$? Round to the nearest thousandth, if necessary.

$x =$

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DIFFERENTIATED RESOURCES

I = Intervention **O** = On-Level **A** = Advanced

Reteach to Build Understanding **I**

Provides scaffolded reteaching for the key lesson concepts.

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6-3 Reteach to Build Understanding

Logarithms

1. The equation $y = \log_a x$ is written in logarithmic form and the equation $y = a^x$ is written in exponential form. But the two are connected in a really straightforward way. Identify the parts of the exponential form. Then, using the information in the diagram at the top of the table below, fill in for the logarithmic form and solve.

Exponential form	Logarithmic form (log base a result)	Solve (if applicable)
$4^x = 64$	$\log_4 64 = x$	NA
Exponent: 4		
Base: 4		
Result: 64		
$10^{x-1} = 1,000$	$\log_{10} 1,000 = x - 1$	$x = 4$
Exponent: $x - 1$		
Base: 10		
Result: 1,000		
$10^x = x$	$\log_{10} x = 2$	$x = 216$
Exponent: x		
Base: 10		
Result: x		

2. A bacterial culture starts with 100 bacteria and the number multiplies by ten every day according to the formula $B = 100(10)^t$, where B is the number of bacteria present after t days. Erin is trying to find the value of t when the number of bacteria is 2,500. Find and correct her error in the solution below.

$2,500 = 100(10)^t$
 $25 = 10^t$
 $\log_{10} 25 = t$ **X** $t = \log 25$
 $t = \frac{1}{10}$ **X** $t \approx 1.4$ days
 $t = 2.5$ **X**

Additional Practice **I O**

Provides extra practice for each lesson.

MathXL® for School

Name _____

6-3 Additional Practice

Logarithms

Complete the table.

Exercise	Exponential form	Logarithmic form
1.	$x^2 = 16$	$\log_2 x = 3$
2.	$x = 5^{2x}$	$\log_5 x = 2$
3.	$4^{-x} = \frac{1}{256}$	$\log_4 \frac{1}{256} = -x$
4.	$x = \frac{1}{100}$	$\log_{10} x = -2$
5.	$100^x = 1$	$\log_{100} 1 = 0$
6.	$32 = \sqrt{x}$	$\log_2 5 = \frac{1}{2}$

Solve the equation for x . Show your work.

7. $2 + \log_2 x = 3$ $\log_2 x = 1$ $x = 2$
 8. $4^{x+2} - 16 = 60$ $4^{x+2} = 76$ $\log_4 76 = x + 2$ $x \approx 1.12$
 9. $2 \ln(x - 5) = 25$ $\ln(x - 5) = 12.5$ $x - 5 = e^{12.5}$ $x = e^{12.5} + 5$

Evaluate each logarithmic expression.

10. $\log_2 \frac{1}{2}$ -1
 11. $\log_8 8$ 1
 12. $\log_5 (-10)$ **No solution**
 13. $\ln(-4)$ **No solution**
 14. $\ln e^3$ 3
 15. $\log 150$ ≈ 2.176

16. Detham invests \$5,000 in a savings account that earns 6% annual interest, compounded continuously. How long will it take to double his money?
about 12 years
 17. Compare the following values and determine which one is greater. Explain.
 $\log_{0.5} 6$ and $\log_{0.5} 4$
 $\log_{0.5} 6 \approx -2.58$ $\log_{0.5} 4 \approx -2$
So $\log_{0.5} 4$ is greater than $\log_{0.5} 6$.

Enrichment **O A**

Presents engaging problems and activities that extend the lesson concepts.

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Name _____

6-3 Enrichment

Logarithms

The following puzzle has been separated into twelve sections.

- Each section has three squares. One square has a number, one has a letter, and one has a number in a circle.
- The circled number in each section tells you the sum of the two missing numbers in the empty squares. All missing numbers are natural numbers: 1, 2, 3, ...
- For example, look at the section marked by (2). The sum of the two missing numbers in the empty squares is 20.
- Complete each equation with numbers that meet the requirements for each given section, and then place them in the puzzle.

A. $\log_2 16 = 4$ B. $\log_3 27 = 3$
 C. $\log_4 64 = 2$ D. $\log_5 625 = 4$
 E. $\log 10 = 1$ F. $\log 16 = 1$
 G. $\log_8 81 = 2$ H. $\log_2 32 = 5$
 I. $\log 144 = 2$ J. $\log 4 = 0.64$
 K. $\log 11 = 1$ L. $\log 216 = 3$

Mathematical Literacy and Vocabulary **I O**

Helps students develop and reinforce understanding of key terms and concepts.

Name _____

6-3 Mathematical Literacy and Vocabulary

Logarithms

Example: Find the inverse of the exponential $y = 10^x - 2$. Justify your work.

Explain	Work	Justify
First, write the equation.	$y = 10^x - 2$	Original equation
Second, add 2 to each side to isolate the exponential.	$y + 2 = 10^x$	Add 2 to each side.
Then, write in logarithmic form.	$x = \log_{10}(y + 2)$	Write in logarithmic form. The result is x with a given base 10 equals the exponent x .
Next, switch x and y .	$y = \log_{10}(x + 2)$	Switch x and y to find the inverse of the exponential x .
Finally, write in common logarithmic form.	$y = \log x + 2$	Write in common logarithmic form.

Solution: $y = \log x + 2$

What is the solution of the equation $\log_5(x - 2) = 2$? Justify your work.

Explain	Work	Justify
First, write the equation.	$\log_5(x - 2) = 2$	Original equation
Second, write in exponential form.	$10^2 = 5x - 2$	Write in exponential form. The base is 10, the exponent is 2, and the result is $5x - 2$.
Then, add 2 to each side.	$100 + 2 = 5x$	Add 2 to each side.
Next, divide each side by 5.	$\frac{102}{5} = x$	Divide each side by 5.
Finally, solve for x .	$x = 20.4$	Simplify to solve for x .

Solution: $x = 20.4$

Digital Differentiated Resources

The Reteach to Build Understanding, Additional Practice, and Enrichment activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.

